

MANASHA SRINITHI G

Electronics & Communication Engineering | Embedded Systems | Firmware | PCB Design | IoT
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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student focused on Embedded Systems, Firmware, PCB Design, IoT, and Hardware R&D.; Hands-on experience with C programming, bare-metal development, STM32 HAL, RTOS concepts, ESP32, Arduino, Raspberry Pi, sensor interfacing, and hardware-software integration. Practical PCB development experience using EasyEDA and KiCad, including schematic design, component selection, placement, routing, ERC/DRC, Gerber generation, and 2D/3D visualization. Additional exposure to Python/OpenCV-based computer vision and IoT development.

TECHNICAL SKILLS

Programming: C, Python, JavaScript

Microcontrollers & Platforms: STM32F103C6T6A, ESP32, ESP32-S3, Arduino, Arduino Nano, Raspberry Pi

Embedded: Bare-Metal Programming, STM32 HAL, RTOS, GPIO, ADC, PWM, UART, I²C, External Interrupts, Task Scheduling, Task Priority, Preemption, Semaphores, Queues, Inter-Task Communication

Hardware & Sensors: IR Sensor, MQ-Series Gas Sensors, LM35, PIR, LDR, Potentiometer, MPU6050, Relay, LED/GPIO Control

PCB & Hardware Design: Schematic Design, PCB Layout, Component Selection, Footprint Assignment, Component Placement, Routing, Copper Pour, Ground Plane, ERC, DRC, Gerber, BOM, 2D/3D PCB Visualization

Tools: EasyEDA, KiCad, Proteus, Wokwi, Tinkercad, ThingSpeak

IoT: ESP32 Wi-Fi, HTTP, IP-Based Device Control, Web-Based Monitoring, ThingSpeak

Computer Vision: Python, OpenCV, Image Processing, Webcam/Camera Processing, Grayscale Conversion, Image Thresholding, Image Resizing

PROFESSIONAL EXPERIENCE

KORVIL TECHNOLOGY | Hardware System Design (PCB) Intern | [Month Year - Month Year]

- Designed PCB-based hardware systems using EasyEDA and KiCad, covering schematic development, component/footprint selection, placement, routing, ERC/DRC validation, Gerber generation, and 2D/3D visualization.
- Developed and analyzed hardware designs involving ESP32, Arduino Nano, LM35, MPU6050, BC547 transistor circuits, relays, and USB-UART interfaces.
- Configured and tested Raspberry Pi 4 Model B with Raspberry Pi OS and Camera Module, including camera hardware integration, live video capture, and Python/OpenCV-based video verification.
- Performed initial image-processing and camera-based computer vision experimentation using the Raspberry Pi platform for SmartCare Sentinel; pose detection and AI-based activity recognition remain under development.
- Contributed to R&D; product analysis, field research, technical documentation, and product-development activities, including interactions with elderly individuals, caregivers, family members, and healthcare professionals.

INETZ TECHNOLOGIES PRIVATE LTD. | Embedded Systems Intern | [Month Year - Month Year]

- Developed an ESP32-based IR sensor object-detection system with real-time detection status displayed through a web interface.
- Implemented Arduino-based MQ-series gas detection with threshold-based LED alerting.
- Implemented ESP32 Wi-Fi/IP-based control for remotely operating LEDs and hardware components over a local network.
- Gained practical experience in sensor interfacing, microcontroller programming, IoT communication, and hardware-software integration.

NOVITECH R&D; PVT. LTD. | Data Analytics, Artificial Intelligence & Machine Learning Intern | 01 Aug 2025 - 30 Aug 2025

- Gained practical exposure to Data Analytics, Artificial Intelligence, and Machine Learning concepts through technical learning and real-time problem-solving activities.

NOVITECH R&D; PVT. LTD. | Full Stack & UI/UX Intern | Feb 2026 - Mar 2026

- Worked on web development and UI design, with focus on developing user-friendly interfaces.

BSNL MADURAI SSA | Inplant Trainee | 1 Week

- Gained practical exposure to telecommunications infrastructure, network switching, transmission, and communication systems, connecting ECE concepts with industry applications.

ZYNOVIX | Co-Founder / Freelance Team Member

- Contributed to a student-led freelancing initiative delivering IoT/AI projects and website, portfolio, and design services.
- Supported client communication, project development, testing, and deployment, including successful delivery of a confidential client IoT project.

SELECTED PROJECTS

SmartCare Sentinel — Intelligent Safety & Emergency Detection System | Raspberry Pi 4 | Camera Module | Python | OpenCV | Image Processing

- Developed the Raspberry Pi hardware foundation with Raspberry Pi OS configuration, camera integration, live capture, and OpenCV-based video verification.
- Performed initial image-processing experimentation for a healthcare/elderly-care application; pose detection and AI-based activity recognition are ongoing development phases.
- Incorporated user research and field interactions to understand elderly safety, medication-management, and independent-living challenges.

Smart Safety Helmet with Ignition Control & Alcohol Detection System | Team InnovaTech | Science Expo 2026

- Developed and presented an embedded smart-safety application focused on alcohol detection and ignition control.

Smart Safety Ring — 15-in-1 Wearable

- Developed a safety and health-monitoring wearable concept incorporating SOS alerting and stress tracking.

ESP32 IR Sensor Object Detection System | ESP32 | IR Sensor | Wi-Fi | Web Interface

- Developed an IoT-based object-detection system using ESP32 and an IR sensor, with Wi-Fi connectivity, web-based monitoring, and real-time detection status.

EMBEDDED SYSTEMS & RTOS PRACTICAL TRAINING

Bare-Metal & STM32 HAL: GPIO, BSRR, LED control, LED chaser, push-button control, external interrupts, PWM, UART, ADC, potentiometer, LM35, PIR, serial monitoring, STM32F103C6T6A, Proteus simulation.

RTOS: Task creation, multiple tasks, task priorities, scheduling, preemption, semaphores, queues, and sender/receiver task communication.

IoT / Wokwi: ESP32-S3, HTTP LED control, PWM, ADC, potentiometer-based brightness control, LDR, temperature sensing, relay control, ThingSpeak, and 5 simulations.

PCB DESIGN PRACTICAL TRAINING

Korvil Technology: EasyEDA, KiCad, component libraries, symbols, footprints, schematic-to-PCB conversion, component placement, routing, copper pour, ground plane, ERC, DRC, Gerber generation, BOM, 2D/3D PCB visualization, and manufacturing concepts.

CERTIFICATIONS & TECHNICAL TRAINING

- PCB Design Foundation Course — Upskill/Gyannidhi
- 3-Day IoT & Embedded Systems Training — iBots.in
- C for Beginners — Great Learning Academy
- Python Image Processing & Computer Vision Training
- 30 Days Masterclass in Data Analytics — NoviTech R&D; Pvt. Ltd.

ACHIEVEMENTS

- First Prize — TechMorph, KREATIV'26, Kamaraj College of Engineering and Technology — ₹3,000 cash prize; system architecture, flowchart, pseudocode, and constrained programming.
- Runner-Up — Visualize'26, National-Level 24-Hour Hackathon, Sethu Institute of Technology — Team InnovaTech; Project: PrevAcc.
- Finalist — HackElite'25, SRM Institute of Science and Technology, Vadapalani — Top 50 of 400+ teams; cleared PPT and prototype-development rounds.
- Selected Team — Nimirndhu Nil Ideation Camp 2026-27, EDII-TN / TNYIEDP — selected among 14 teams from 150+ Dindigul registrations.